AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# M348
# Applied Statistical Modelling

## Midterm Examination (MTA)
## Fall Semester 2025/2026

## Date: 18/Nov./2025

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|---|---|
| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 2 Hours |

**Instructions:**

- Total Marks: 30
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.



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## PART 1: True or False Questions [5 marks]

Which of the following statements are true/false?

a) The Total Sum of Squares (TSS) measures the total variation in the response variable.
b) A negative intercept always means the model is invalid.
c) $R^2$ of 0.95 means that 95% of the variation in X is explained by Y.
d) A scatterplot of residuals vs fitted values helps check for non-linearity.
e) Multicollinearity makes coefficient estimates unstable.
f) Forward selection starts with no variables and adds them one by one.
g) Adjusted $R^2$ always decreases when more predictors are added.
h) High bias means the model is too simple to capture the relationship.
i) The null model in regression contains only the intercept.
j) For left-skewed data, we move up the ladder of powers (e.g., $x^2$, $x^3$).

## PART 2: Multiple Linear Regression for the bike dataset [13 marks]

ALL QUESTIONS ARE REQUIRED

The Bike dataset records the hourly and daily bike rental counts in the Capital Bikeshare system during 2011–2012, along with relevant weather and seasonal information.

Key variables include:

- season (categorical: 1-4): Season indicator (1 = winter, 2 = spring, 3 = summer, 4 = fall)
- temp: Normalized temperature in Celsius UCI Machine Learning Repository
- atemp : "Feels like" temperature, normalized in Celsius UCI Machine Learning Repository
- hum: Normalized humidity (scaled) UCI Machine Learning Repository
- windspeed : Normalized wind speed UCI Machine Learning Repository
- casual : Count of casual (non-registered) bike users UCI Machine Learning Repository
- registered: Count of registered bike users UCI Machine Learning Repository
- cnt : Total count of bike rentals (casual + registered)

You are asked to fit a multiple linear regression model where the response variable is cnt and the explanatory variables are temp, atemp, hum, windspeed, casual and registered.



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```
Residuals:
      Min        1Q    Median        3Q       Max
-3.899e-11 -3.200e-13 -9.000e-14  1.450e-13  4.123e-11

Coefficients:
             Estimate Std. Error    t value Pr(>|t|)
(Intercept) 6.379e-12  6.964e-13  9.160e+00  < 2e-16 ***
temp        5.593e-12  4.147e-12  1.349e+00  0.17789
atemp       1.220e-11  4.691e-12  2.602e+00  0.00946 **
hum        -8.426e-12  7.306e-13 -1.153e+01  < 2e-16 ***
windspeed  -9.682e-12  1.345e-12 -7.201e+00  1.5e-12 ***
casual      1.000e+00  1.709e-16  5.850e+15  < 2e-16 ***
registered  1.000e+00  7.631e-17  1.310e+16  < 2e-16 ***
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.582e-12 on 724 degrees of freedom
Multiple R-squared:      1, Adjusted R-squared:      1
F-statistic: 6.849e+31 on 6 and 724 DF,  p-value: < 2.2e-16
```

1. Using the results obtained above, what is the fitted equation for this model?
2. Interpret the coefficient of atemp in the model. What does this value mean in the context of bike rentals?
3. What does the adjusted $R^2 = 1$ tell us about the model, and why is adjusted $R^2$ more reliable than multiple $R^2$ when comparing models?
4. The residual standard error (RSE) is very close to 0. Suppose a previous simple regression model (with only temp as predictor) had a much higher RSE. What does this reduction suggest about the model's improvement?
5. How could multicollinearity among predictors (e.g., temp and atemp) affect the interpretation of coefficients? Suggest one method to detect it and one method to address it.
6. Suppose you add more predictors such as season, weekday, and weathersit. Write the R code to perform both forward and backward stepwise regression starting from the full model.
7. Suppose two competing models both have high adjusted $R^2$ but one has a much higher residual standard error. Which model would you choose, and why? What assumptions of simple linear regression should be checked before trusting the model results?
8. If the distribution of the total bike rentals (cnt) shows a long right tail, which type of ladder-of-powers transformation would be most appropriate, and why?
9. If stepwise regression selects a model where predictors appear in log-transformed form (e.g., log(temp), log(hum)), what does this imply about the relationship between those predictors and bike rentals?
10. If the transformed model shows lower residual standard error and residuals closer to normality, should this model automatically be chosen over the untransformed model? Why or why not?

## PART 3: Simple Regression with a factor (Bike dataset) [12 marks]

ALL QUESTIONS ARE REQUIRED



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You are asked to fit a simple linear regression model where the response variable is cnt and the explanatory variable is a factor season.

```
Residuals:
    Min      1Q  Median      3Q     Max
-4706.2 -1065.6  -181.3  1219.2  5231.9

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    2604.1      116.6   22.33   <2e-16 ***
seasonSpring   2388.2      164.2   14.54   <2e-16 ***
seasonSummer   3040.2      163.4   18.61   <2e-16 ***
seasonFall     2124.0      165.6   12.83   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1569 on 727 degrees of freedom
Multiple R-squared:  0.347,     Adjusted R-squared:  0.3443
F-statistic: 128.8 on 3 and 727 DF,  p-value: < 2.2e-16
```

1. In the model cnt ~ season, identify the dependent and independent variables.
2. Based on the output, which season has R taken as the baseline (reference category)?
3. Interpret the estimated value of the intercept (2604.1). What does it represent in terms of bike rentals?
4. Interpret the estimated coefficient for Spring (2388.2). What does this mean compared to Winter?
5. Write the fitted regression equation using the coefficients from the summary.
6. Use the model to predict the expected number of rentals in Fall.
7. Why is it important to check residual plots (e.g., for normality and constant variance) after fitting this regression model?
8. If another variable such as temp were added to the model, why might the Adjusted R-squared be more useful than the regular R-squared for evaluating the model?
9. Write the null and alternative hypotheses for the overall ANOVA F-test of this model.
10. Which function in R would you use to perform this ANOVA test?

End of questions



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